

GURLAL SINGH

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PROFESSIONAL SUMMARY

Computer Science (AI) undergraduate at Bennett University with hands-on experience building agentic AI systems, LLM-powered applications, and full-stack products using Python, Node.js, React.js, and Next.js. Skilled in RAG pipelines, LangChain/LangGraph agent orchestration, prompt engineering, browser automation, and REST API development, backed by a published IEEE research paper in deep learning. Strong foundation in Data Structures & Algorithms, OOPs, DBMS, and Software Engineering, with a track record of shipping production-ready AI applications that automate real business workflows.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, SQL

AI & Generative AI: AI Agents / Agentic AI, LLMs, RAG, LangChain, LangGraph, OpenAI API, OpenRouter, Prompt Engineering, Machine Learning, Deep Learning, NLP, Computer Vision

Full-Stack Development: React.js, Next.js, Node.js, Express.js, NestJS, FastAPI, REST APIs, JWT Authentication, Playwright

Databases: PostgreSQL, MongoDB, Firebase, SQLAlchemy, Supabase

Data Science: Pandas, NumPy, Data Analysis, Data Preprocessing, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Tools & Platforms: Git, GitHub, Docker, Postman

Core CS Foundations: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, System Design

EDUCATION

Bennett University (Times Group), Greater Noida, Uttar Pradesh

B.Tech, Computer Science — Specialization in Artificial Intelligence | 2023–2027 | CGPA: 8.22/10

WORK EXPERIENCE

AI Research Intern — Bennett University

June 2025 – Sept 2025

- Compared and benchmarked multiple deep learning architectures for medical-imaging and autism-detection research through structured literature review, enabling the team to shortlist the highest-performing model configurations.
- Built end-to-end deep learning pipelines in Python covering data preprocessing, feature engineering, model training, and hyperparameter tuning on medical imaging datasets to improve classification accuracy.
- Authored technical documentation and research reports summarizing experimental findings for faculty and stakeholders, contributing directly to a paper accepted at an IEEE international conference.
- Synthesized technical papers and research findings into concise summary reports, accelerating stakeholder review and faster decisions on next experimental steps.

PROJECTS

CV Tailor — Autonomous Multi-Agent AI Resume Platform

React.js, FastAPI, LangGraph, LangChain, OpenAI, PostgreSQL, SQLAlchemy, JWT, Supabase, PyMuPDF, Docker | github.com/thegurlalsingh/cv_tailor

- Designed an autonomous multi-agent AI platform orchestrating Strategist, Writer, Auditor, ATS Evaluator, and Revision Planner agents to iteratively tailor resumes to job descriptions and improve ATS match scores.
- Architected scalable backend services with FastAPI, PostgreSQL, and JWT authentication, and implemented real-time streaming of agent execution via Server-Sent Events for live visibility into AI reasoning.
- Integrated LLM orchestration, PDF parsing (PyMuPDF), and agent-based fact verification, eliminating hallucinated content and improving factual accuracy in generated resumes.

Botfolio — AI-Powered Recruitment Platform

React.js, Next.js, Node.js, NestJS, PostgreSQL, JWT, REST APIs, Qwen3 LLM, Docker | github.com/thegurlalsingh/botfolio

- Built a full-stack AI-powered recruitment platform automating the end-to-end candidate assessment lifecycle, with secure REST APIs using JWT-based authentication and role-based authorization.
- Developed a role-based interview generation engine using prompt engineering with Qwen3 LLMs, dynamically evaluating candidate responses and improving evaluation consistency.

- Engineered AI-assisted workflow automation for candidate screening, ranking, and reporting, accelerating recruiter time-to-decision at scale.

Creator Discovery Automation — AI Influencer Outreach Platform

React.js, FastAPI, Playwright, Supabase/PostgreSQL, OpenRouter AI, Resend | github.com/thegurlalsingh/creator_discovery_automation

- Built an end-to-end AI-powered platform that discovers Instagram micro-influencers, verifies profile metrics via Playwright automation, and calculates engagement rate from recent content.
- Integrated an LLM pipeline via OpenRouter to classify creator niche and content themes, and auto-generate personalized cold emails and Instagram DMs grounded in each creator's actual content.
- Designed a relational Supabase/PostgreSQL schema tracking creators, content, and outreach status, with automated email delivery and status tracking through Resend.

Scholarship Extractor — Automated Scholarship Intelligence Pipeline

Python, FastAPI, BeautifulSoup, Supabase PostgreSQL, pg_cron, React | github.com/thegurlalsingh/scholarship_extractor

- Built a discover-crawl-extract-verify-score-monitor pipeline that runs 280+ targeted search queries across government, university, CSR, and NGO sources to surface authentic scholarships for Indian students.
- Implemented a deterministic-first extraction engine with LLM fallback and an evidence-based confidence scoring system to flag verified vs. review-required listings, minimizing hallucinated data.
- Automated continuous change detection with Supabase pg_cron, tracking field-level history and computing live status (active/expiring/expired) surfaced through a React dashboard.

PUBLICATIONS

"Comparative Analysis of Various Deep Learning Models for Autism Detection Using Grid Search"

IEEE Xplore International Conference, 2025

- Performed comparative analysis of deep learning architectures and feature engineering strategies for autism detection, applying Grid Search hyperparameter optimization to improve model accuracy and contribute to the paper's acceptance.

LEADERSHIP

Vice President, Operations — National Service Scheme (NSS), Bennett University

2025–2026

- Led a cross-functional team of student volunteers across university-wide social impact initiatives, overseeing operations, stakeholder management, and event planning that increased volunteer participation.

ACHIEVEMENTS

- Published an IEEE conference paper on deep learning, hyperparameter optimization, and medical AI applications.
- Solved 600+ LeetCode problems with a contest rating of 1500+, demonstrating strong data structures and algorithms proficiency.
- Ranked 88th out of 650+ teams in the Smart BU Hackathon.